

Group Project Proposal

Course Code: CSC204

Course Title: Database Systems

Semester: _____ IV _____

Group Members

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Project Details

Project Title: _____ Purrfect Care _____

Proposed Database Name: _____ Supabase (Postgresql) _____

1. Project Overview

Purrfect Care is an all-in-one cat healthcare and lifestyle platform designed to bridge the gap between cat owners, veterinarians, and pet stores. The platform addresses a fragmented ecosystem where cat owners must juggle separate apps for booking vet appointments, getting prescriptions, buying supplies, and tracking their cat's health history.

Main Objectives:

- Provide a centralized platform for managing a cat's complete health life cycle
- Enable direct, real-time communication between cat owners and verified veterinarians
- Automate prescription workflows and medical record management
- Offer an integrated e-commerce store for cat products
- Incorporate an AI companion for diagnosing common cat illnesses

2. Proposed Solution

Purrfect Care uses a relational database to model the complex, interconnected relationships between users, cats, vets, hospitals, and stores. The system solves the problem through:

- Centralized Patient Records: Each cat has a persistent medical record and full patient history (diagnoses, vaccinations, prescriptions, surgeries).
- Appointment Scheduling Engine: Slot-based booking system prevents double-booking; stateful appointments track the full life cycle.
- Prescription-to-Store Pipeline: Vets issue digital prescriptions that reference the medicines catalog; prescription status is tracked and shareable via chat.

- Geo-aware Discovery: PostGIS extension enables proximity-based hospital and store search using latitude/longitude coordinates.
- Role-Based Access Control: Five distinct user roles (cat owner, vet, hospital admin, store owner, admin) each with scoped permissions enforced via Row-Level Security.

3. Main Entities and Attributes

- UserProfile: user_id, name, phone, role, latitude, longitude, preferences
- Cat: owner_id, name, breed_id, age_months, weight_kg, gender, is_neutered
- CatBreed: name, origin_country, avg_lifespan_years, common_health_issues
- MedicalRecord: cat_id, allergies, existing_conditions, vaccination_status
- PatientHistory: cat_id, entry_type, description, appointment_id, vet_id
- Hospital: name, address, city, latitude, longitude, operating_hours, rating
- HospitalService: hospital_id, name, category, price, duration_minutes
- Vet: user_id, hospital_id, license_number, specialization, experience_years
- AppointmentSlot: vet_id, hospital_id, slot_date, start_time, end_time, is_booked
- Appointment: user_id, cat_id, vet_id, service_id, slot_id, status
- Medicine: name, generic_name, dosage_form, requires_prescription • Prescription: cat_id, vet_id, medicine_id, dosage, frequency, duration_days
- ChatRoom: user_id, vet_id, last_message_at, unread_user, unread_vet
- Message: chat_room_id, sender_id, content, message_type
- CatStore: owner_user_id, name, city, delivery_fee, latitude, longitude
- Product: store_id, category_id, name, price, stock_quantity, brand
- Order: user_id, store_id, total, status, delivery_address
- Review: user_id, hospital_id/store_id/vet_id, rating, comment
- Payment: user_id, appointment_id/order_id, amount, stripe_payment_id, status

4. Expected Outcomes

After completing this project, the team expects to:

- Deliver a fully functional platform with role-based dashboards for cat owners, vets, hospital admins, and store owners.
- Implement a real-time chat system between owners and vets with prescription sharing via in-chat messages.

- Achieve end-to-end appointment flow: browse, book, pay, consult, receive prescription, and re-order medicine.
- Build a geo-aware discovery layer allowing users to find nearby hospitals and stores by proximity.
- Enable rich analytics for admins: appointment trends, top-rated vets, revenue tracking across stores and hospitals.
- Integrate an AI companion powered by OpenAI to provide first-level triage and guidance for common cat health concerns.

5. Tools and Technologies

- Frontend: React (Vite), JavaScript, Axios, React Query, React Router
- Backend: Python, FastAPI, Pydantic
- Database (DBMS): PostgreSQL via Supabase (hosted, managed Postgres)
- DB Extensions: uuid-oss (UUID generation), PostGIS (geospatial queries)
- Authentication: Supabase Auth (JWT-based) with Row-Level Security
- Payments: Stripe API
- Real-time: Supabase Realtime (WebSocket-based live chat & notifications)
- AI / ML: Google Gemini API (Free Tier) or Groq API (Llama 3) for the AI companion feature
- API Design: OpenAPI 3.0 / REST